



GENERATIVE AI FOR TRADITIONAL FORM CONVERTER

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A PROJECT SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR
THE DEGREE OF BACHELOR OF ENGINEERING (COMPUTER ENGINEERING)
FACULTY OF ENGINEERING
KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI
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Abstract

The Generative AI for Traditional Form Converter is a project developed through a web application called PaperlessTransform Application, aimed at solving the problem of time-consuming processes involved in converting PDF forms into web applications. This web application leverages artificial intelligence to analyze data types of input labels. As the demand for form conversion continues to increase, system developers are required to analyze a PDF forms structure, design corresponding database structures, create web application interfaces, and develop new systems. These tasks result in significantly increased work time and workload, causing personnel to spend time inefficiently. The goal of this project is to reduce the burden on system developers by automating parts of this workflow. The application is designed to process a PDF file, detect an input fields using parser, generate an input field data type using AI-powered text analysis, and generate web-based forms accordingly. In addition to form field detection, the system can store form data in a structured format suitable for web deployment, enabling a smoother transition from traditional documents to digital interfaces. The project team has focused on developing a robust system that not only identifies relevant form elements accurately but also simplifies the data handling and form creation process. After testing the web application, results show that it can detect input field within forms and store data at a basic level, demonstrating its practical effectiveness in real-world scenarios. Therefore, it can be concluded that this project addresses the problem of increased work time and inefficiency for system developers, offering an effective tool for automated form conversion and reducing the manual effort required in digitizing traditional forms.

Keywords: Web Application / Database Design

หัวข้อปริญญานิพนธ์	เว็บแอปพลิเคชัน AI สำหรับการแปลงฟอร์มกระดาษเป็นเว็บฟอร์ม
หน่วยกิต	3
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บทคัดย่อ

Generative AI for Traditional Form Converter เป็นโครงการที่จัดทำขึ้นผ่านการพัฒนาผ่านเว็บแอปพลิเคชันในชื่อ PaperlessTransform Application เพื่อแก้ไขปัญหาการใช้ระยะเวลานานในการแปลงแบบฟอร์มกระดาษเป็นรูปแบบเว็บแอปพลิเคชัน โดยการพัฒนาเว็บแอปพลิเคชันนี้ได้มีการใช้ประยุกต์ใช้ปัญญาประดิษฐ์สำหรับการวิเคราะห์เกี่ยวกับประเภทของข้อมูลของคำถาม ตามความต้องการที่เพิ่มขึ้นของการแปลงแบบฟอร์ม ดังนั้นนักพัฒนาระบบจึงจำเป็นต้องวิเคราะห์แบบฟอร์มและออกแบบระบบฐานข้อมูลพร้อมทั้งการออกแบบหน้าเว็บแอปพลิเคชัน รวมไปถึงการพัฒนาระบบขึ้นมาใหม่ จึงส่งผลให้ต้องใช้ระยะเวลาในการทำงานที่เพิ่มขึ้น นอกจากนี้ นักพัฒนาระบบต้องเผชิญกับปัญหาภาระงานที่มากขึ้น ส่งผลให้บุคลากรใช้เวลาในการทำงานอย่างไม่มีประสิทธิภาพ โดยโครงการของเรามุ่งเน้นการพัฒนาเว็บแอปพลิเคชันที่สามารถแปลงเอกสารในรูปแบบไฟล์อิเล็กทรอนิกส์ให้เป็นรูปแบบของเว็บแอปพลิเคชัน โดยนำข้อความจากไฟล์อิเล็กทรอนิกส์ดังกล่าวมาประมวลผลในการตรวจจับคำถามในรูปแบบฟอร์ม ทางคณะผู้จัดทำโครงการมีการเน้นการพัฒนาเว็บแอปพลิเคชันที่มีความสามารถในการตรวจจับคำถามและความสามารถในการเก็บข้อมูลของเว็บฟอร์ม โดยมีวัตถุประสงค์เพื่อลดภาระของนักพัฒนาระบบ โดยผลลัพธ์หลังจากมีการทดลองใช้เว็บแอปพลิเคชันดังกล่าวในการทำงานแสดงให้เห็นว่าเว็บแอปพลิเคชันสามารถตรวจจับคำถามในแบบฟอร์มและเก็บข้อมูลได้ในระดับที่น่าพึงพอใจ ดังนั้นสรุปได้ว่าโครงการสามารถแก้ไขปัญหาการใช้ระยะเวลาในการทำงานที่เพิ่มขึ้น ของนักพัฒนาระบบได้อย่างมีนัยสำคัญ

คำสำคัญ: เว็บแอปพลิเคชัน / ปัญญาประดิษฐ์ / ออกแบบระบบฐานข้อมูล / ไฟล์อิเล็กทรอนิกส์ / นักพัฒนาระบบ

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CONTENTS

	PAGE
ABSTRACT	ii
THAI ABSTRACT	iii
ACKNOWLEDGMENTS	iv
CONTENTS	v
LIST OF TABLES	viii
LIST OF FIGURES	ix
LIST OF SYMBOLS	x
LIST OF TECHNICAL VOCABULARY AND ABBREVIATIONS	xi
 CHAPTER	
1. INTRODUCTION	1
1.1 Problem Statement	1
1.2 Objectives	1
1.3 Scope of Work	1
1.4 Limitation of Project	1
1.5 Project Schedule	2
1.6 Expected Outcomes	3
 2. BACKGROUND THEORY AND RELATED RESEARCH	4
2.1 Introduction and Background	4
2.1.1 Introduction	4
2.1.2 Background	4
2.2 Theory and Core Concepts	4
2.2.1 Artificial Intelligence (AI)	4
2.2.2 Machine Learning (ML)	4
2.2.3 Natural Language Processing (NLP)	5
2.2.4 Deep Learning	5
2.2.5 Generative AI	6
2.2.5.1 Types of Transformer models	6
2.2.6 Image Processing	6
2.2.6.1 Image Acquisition	6
2.2.6.2 Preprocessing	6
2.2.6.3 Segmentation	7
2.2.6.4 Feature Extraction	7
2.2.6.5 Object Recognition	7
2.2.7 Optical Character Recognition (OCR)	7
2.2.8 OCR Architecture	8
2.2.9 Portable Document Format	8
2.2.9.1 PDF Categories	8
2.2.9.2 PDF File Structure	9
2.2.10 Text Parser	9
2.2.11 No Language Left Behind (NLLB)	9
2.2.11.1 Model Architecture and Training	9
2.2.12 Security of Web Form	11

2.3	Languages and technologies	12
2.3.1	Web Development Language	12
2.3.1.1	HTML	12
2.3.1.2	TypeScript	12
2.3.2	Front-end Technology	12
2.3.2.1	Vue.js	12
2.3.2.2	TailwindCSS	12
2.3.2.3	Formkit	12
2.3.3	Backend Technology	12
2.3.3.1	FastAPI	12
2.3.3.2	Pydantic	13
2.3.3.3	SQLModel	13
2.3.3.4	PostgresSQL	13
2.3.4	Document Processing	13
2.3.4.1	Python	13
2.3.4.2	PyMuPDF	13
2.3.4.3	NLLB-200	13
2.3.4.4	PyTorch	13
2.3.4.5	LLAMA	13
2.4	Competing solutions	14
2.4.1	Microsoft Azure Document Intelligence	14
3.	DESIGN AND METHODOLOGY	15
3.1	Project Functionality	15
3.1.1	System Requirements	15
3.1.2	Feature List	15
3.1.2.1	Registration and Authentication System	15
3.1.2.2	Form Layout Analysis Feature	15
3.1.2.3	Form Schema generator Feature	15
3.1.2.4	Web Application Form	15
3.2	Use Case Diagram	16
3.3	Use Case Narrative	16
3.3.1	Autentication	16
3.3.2	Register	17
3.3.3	Create a form	17
3.3.4	Edit a form	17
3.3.5	View a form data	18
3.3.6	Fill up the form	18
3.4	Activity Diagram	19
3.4.1	Form creator	19
3.4.2	Signed-in user	19
3.4.3	Non Signed-in user	19
3.5	System Architecture	20
3.6	System Workflows	20
3.6.1	Upload a file	20
3.6.2	Extracting text	20
3.6.3	Text Parse	21
3.6.4	Translation	21
3.6.5	Data Type Generator	21
3.6.6	Form Schema Generator	21

3.7	System Design	21
3.7.1	Text Extraction Design	21
3.7.1.1	PDF Text Extraction	22
3.7.1.2	Optical Character Recognition Text Extraction	22
3.7.2	Text Parse Design	23
3.7.3	Data Type Generator Design	23
3.8	Database Design	24
3.9	User Interface Design	25
3.9.1	Login Page	25
3.9.2	Create Your Account	25
3.9.3	Edit Profile	25
3.9.4	Forgot Password	26
3.9.5	Dashboard	27
3.9.6	Edit Form	28
3.9.7	Form Page	28
4.	IMPLEMENTATION RESULT	30
4.1	Introduction	30
4.2	Data Collection and Data Categorized	30
4.2.1	Data Processing	32
4.2.2	Text Parser Result	33
4.2.3	Translation Thai to English	34
4.2.4	Data Type Analyzer Result	34
4.3	Web Application Implementation	35
4.3.1	Front-end development	35
4.3.2	Back-end development	37
5.	CONCLUSIONS	40
5.1	Introduction	40
5.2	Workloads	40
5.3	Problems and Solutions	41
5.3.1	Lack of Optical Character Recognition	41
5.3.2	Lack of PDF Extraction	41
5.3.3	Lack of Tabular form support	42
5.3.4	Language model for generating answers	42
5.4	Future Works	42
	REFERENCES	44

LIST OF TABLES

TABLE	PAGE
4.1 Page Route API Specification	38
4.2 Authentication Routes API Specification	38
4.3 Form and User Routes	39
5.1 Workload and Progress Summary	41

LIST OF FIGURES

FIGURE	PAGE
1.1 Schedule for first semester	2
1.2 Schedule for second semester	3
2.1 Natural Network from Qtravel.ai .	5
2.2 Image Segmentation Explained from builtin.com .	7
2.3 OCR Process from researchgate.net .	8
2.4 Microsoft Azure Document Intelligence from Learn.microsoft.com .	14
3.1 Use case diagram	16
3.2 Activity Diagram	19
3.3 System Architecture	20
3.4 System Workflow Diagram	21
3.5 Text Parse Flowchart	23
3.6 ER Diagram	24
3.7 Login Page	25
3.8 Create Account Page	25
3.9 Edit Profile Page	26
3.10 Forgot Password Page	26
3.11 Forgot Password Page	27
3.12 Reset Password Page	27
3.13 Dashboard Page	28
3.14 Edit Form Page	28
3.15 Form Page	29
4.1 Example of dot-style forms	30
4.2 Example of line-style forms	31
4.3 Example of tabular-style forms	32
4.4 OCR Text Extraction Result	33
4.5 PDF Text Extraction Result	33
4.6 Form Parse Result	34
4.7 Translation Result	34
4.8 Data Type Analyzer Result	35
4.9 Implemented Login Page	36
4.10 Implemented Signup Page	36
4.11 Implemented Forgot Passowrd	36
4.12 Implemented Dashboard Page	37

LIST OF SYMBOLS**SYMBOL****UNIT**

LIST OF TECHNICAL VOCABULARY AND ABBREVIATIONS

AI	=	Artificial Intelligence
API	=	Application Programming Interface
BERT	=	Bidirectional encoder representations from transformers
CDN	=	Content Delivery Network
CCPA	=	California Consumer Privacy Act
CSRF	=	Cross-Site Request Forgery
CSS	=	Cascading Style Sheets
E2EE	=	End-to-end encryption
EV	=	Extended Validation
GANs	=	Generative Adversarial Networks
GDPR	=	General Data Protection Regulation
GPT	=	Generative Pre-trained Transformer
HTML	=	Hypertext Markup Language
LLAMA	=	Large Language Model Architecture
ML	=	Machine Learning
NLP	=	Natural Language Processing
NLLB	=	No Language Left Behind
OTP	=	One Time Password
PDF	=	Portable Document Format
SQL	=	Structured Query Language
SSL	=	Secure Sockets Layer
TLS	=	Transport Layer Security
UX/UI	=	User Experience and User Interface
VAEs	=	Variational Auto Encode
XSS	=	Cross-Site Scripting

CHAPTER 1 INTRODUCTION

1.1 Problem Statement

In recent year, the world has become more digitized than ever, whether it be using electronic devices for note-taking instead of paper or storing a data in a database rather than use a paper documentation, However, there are still many aspects that have not been transform to be a digital, such as a government organization that still rely on paper documentation or the data forms that have been previously recorded on paper. However, for those that have not been developed to be digital, we were motivated to improve the efficiency of form filling. And we found that the development process from paper-based forms to web forms required developers to do an analysis of the form, create a database, and develop a front-end and backend system. Which causes a developer time to spend and resources of developers. We acknowledge this difficulty and offer a solution that converts paper forms into web form ones so that the system can automatically create a web form by just taking a picture of the form. By decreasing excessive paper use, this development in digitizing form-filling procedures will not only increase efficiency but also lessen the environmental impact and help to slow down global warming.

1.2 Objectives

- To reduce the workload and development process for developers.
- To acquire the knowledge and skills necessary for developing an AI-powered web application
- To acquire proficiency in utilizing a Large language models and adapt its capabilities to suit the requirements of this project.
- To be the secure all data and form management website

1.3 Scope of Work

The scope of this project involves the development of a web application that enables users to upload a form file. The web application will process text extraction using technique called pdf parsing for PDF file and OCR for image. The primary function of the web application includes creating a form, editing a form, deleting a form, and filling a form. The final deliverable of this project will be a web application that allows users to manage the form and view the data, including ensuring data privacy and security measures to protect sensitive information by implementing authentication for the creator and a normal user. The project involves research of text parser for extract input label from a text and the development of generative AI for data type generation, also a Thai language translation to English.

1.4 Limitation of Project

The limitation of the project will addresses a possible constraints and challenges that might affect its scope, execution, or outcomes. the limiting factor are include time, cost and risk etc.

- **OCR Accuracy:**The accuracy of OCR text extraction may vary depending on the quality of the uploaded document, such as low-resolution images, poor lighting, and the project does not support hand-written text.

- **PDF Support:** The project supports text-based PDFs but not image-based ones. Text within scanned documents cannot be extracted due to the lack of OCR support. Additionally, some PDFs may fail during text extraction because they are compressed or contain missing characters.
- **Language Support:** While the system has the ability to translate text, the accuracy and availability of supported languages may be limited, with the project currently supporting Thai and English form only.
- **Required User Reviewing:** After text extraction and layout detection, the system requires users to review and correct the input labels it processed. This limitation exists due to text extraction accuracy and generative AI accuracy.
- **Data Privacy and Security:** Despite the implementation of verification and security measures, there may still be vulnerabilities related to handling sensitive data, which are continuously checked and updated.

1.5 Project Schedule

For the first semester, our project focus on researching and design phase, We have researching all core fundamental concept and define a problem and background of the project. In the design phase, we have design a database design, UX/UI design, architecture design. And this phase also including a Optical Character Recognition proof of concept.

TASK		SEMESTER 1																			
		AUG				SEP				OCT				NOV				DEC			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Proposal	Discussion Project																				
	Write Project Idea																				
	Write Project Report																				
	Make Proposal Presentation																				
Project Planning	Plan Task Schedule																				
	Research Model and AI																				
Learning and Research	Study Color Image Processing																				
	Study Image Compression																				
	Study Morphological Image Processing																				
	Study Representation and Description																				
Design And Data Preparation	Make Use case Diagram																				
	Make Architecture Diagram																				
	Design Database																				
	Design AI Design																				
	Design UX/UI																				
Implementation	Select an Appropriate State of the art AI Model																				
	Select an Appropriate Front End and Back End Framework																				
	Select an Appropriate Security Framework																				
Final Report	Write Final Report																				
	Make Final Presentation																				

Figure 1.1 Schedule for first semester

For the second semester, our project focus on implementing the form extractor and generative AI for the process of detect a form. We also focus on web application development and integrate a web application with the form extractor. also including the testing phase a system evaluation.

		SEMESTER 2																			
TASK		JAN				FEB				MAR				APR				MAY			
		1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4	1	2	3	4
Form Layout Detection	OCR Implementation																				
	Form Layout Identification																				
	Text Error Correction																				
Translation Implement	Generative AI Implementation																				
Web application implementation	Backend																				
	Frontend																				
	Test Case Design																				
	Unit Testing																				
Project Report	Write Final Report																				
	Make Final Presentation																				

Figure 1.2 Schedule for second semester

1.6 Expected Outcomes

This project aims to develop a fully functional web application that able to converting a paper-based form or pdf form into a web-based form by utilizing a generative AI for generate a data type of form label. And the web application should reduce a time developer have spend when they converting a form.

CHAPTER 2 BACKGROUND THEORY AND RELATED RESEARCH

2.1 Introduction and Background

2.1.1 Introduction

This chapter will explain the details of the core concept and the solution planning. Theory and core concepts, languages and technologies, and related research will be discussed in this chapter. First, we will cover a core concept of artificial intelligence, machine learning, natural language processing and image processing etc.. Second, the languages and technologies that we interest in the project including a frontend and backend technology. Lastly, related research and competing solutions that are similar to our project will be in research and competing solutions.

2.1.2 Background

The digital transformation of businesses has been accelerated by the need for faster, more reliable ways to handle data. Although many organizations have begun to adopt digital processes, a significant number still rely on paper forms, which can slow down operations and increase the risk of errors. Manual data entry, in particular, is an inefficient method that often leads to mistakes, misinterpretations, and lost time.

To solve these problems, many organizations are looking for ways to turn paper forms into digital formats automatically. This is where AI comes in. AI tools can be trained to read forms and convert them into digital versions, speeding up the process and reducing errors. By automating this task, businesses can save time, lower costs, and reduce mistakes, allowing employees to focus on more important tasks.

This chapter will explain the main ideas behind the project. It will also discuss the technologies used in the project and look at similar research in the field of form automation.

2.2 Theory and Core Concepts

2.2.1 Artificial Intelligence (AI)

Artificial Intelligence (AI) refers to the study and development of intelligent machines and software that can reason, learn, communicate, and perceive objects, aiming to mimic human-like behavior. Coined by John McCarthy in 1956, AI is a branch of computer science that focuses on enabling computers to perform tasks typically requiring human intelligence, such as problem-solving, perception, and decision-making.

2.2.2 Machine Learning (ML)

Machine Learning is an application of Artificial Intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed. Machine Learning is crucial in building systems that can automatically learn to recognize patterns and improve over time with more data.

Machine Learning Method:

- **Supervised Learning:** A type of machine learning where the model is trained using a labeled dataset. This means the data comes with answers, so the model learns to make predictions or classify information correctly.
- **Unsupervised Learning:** A type of machine learning where the computer learns from data without labels or answers. Instead of being told what to look for, the model tries to find patterns or group similar data points together on its own.

- **Semi-supervised learning:** Like a mix of supervised and unsupervised learning. It uses a small amount of labeled data (with answers) and a large amount of unlabeled data (without answers) to train the model. The labeled data helps guide the model, while the unlabeled data helps it find patterns and improve accuracy.

2.2.3 Natural Language Processing (NLP)

Natural Language Processing (NLP) enables machines to understand, interpret, and generate human language in a valuable way. The process involves several key techniques: Tokenization: Breaking down text into smaller units (tokens), such as words or phrases, which can be analyzed separately. Stop Word Removal: Eliminates common words that provide little informational value, focusing on more meaningful words. Lemmatization and Stemming: Reduces words to their root forms to unify different inflected versions (e.g., "walking" becomes "walk"). Part-of-Speech Tagging: Assigns tags to words based on their grammatical roles (nouns, verbs, adjectives, etc.).

2.2.4 Deep Learning

Deep Learning is a type of machine learning where computers are trained to think and learn like humans by processing large amounts of data. It uses a special kind of program called a neural network, which is inspired by the way our brains work. These networks can figure out patterns and make decisions on their own, such as recognizing faces in photos, translating languages, or even predicting what you might want to buy.

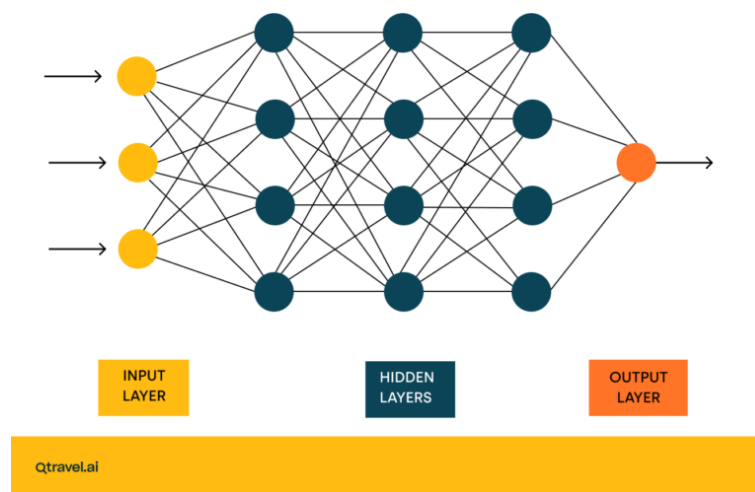


Figure 2.1 Natural Network from [Qtravel.ai](https://qtravel.ai).

2.2.5 Generative AI

Generative AI refers to deep learning models that can generate high-quality text, images, and other content based on trained data. It works by analyzing a lot of data, such as Wikipedia articles or famous paintings, and then using what it learns to produce original outputs similar to the data it studied.

In the past, AI models like Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) helped create realistic images and sounds. Now, newer models like transformers have taken over. Transformers are powerful because they can learn from massive amounts of text without needing people to label everything.

2.2.5.1 Types of Transformer models

1. **Encoder-only models** (such as BERT): Great for understanding and categorizing text but not for generating new content.
2. **Decoder-only models** (such as GPT): Focus on predicting the next word and are good at creating new text.
3. **Encoder-decoder models** (such as T5): Combine both approaches and can handle tasks like summarizing or translating text.

2.2.6 Image Processing

Image processing is the one technique in Computer Vision (CV) that is used to enhance and prepare images for analysis by applying various computational algorithms. In the context of form conversion, image processing plays a critical role in improving the quality of scanned documents or digital images before they undergo text recognition.

2.2.6.1 Image Acquisition

The first step in image processing is acquiring the image, where an image is captured using a camera, scanner, or another device. Then the image is converted into a digital format that can be manipulated by algorithms.

2.2.6.2 Preprocessing

Preprocessing is a fundamental step where the raw image quality is improved to ensure it is ready for more advanced analysis. This involves techniques such as:

Image Enhancement

Image enhancement involves using techniques to improve the visual quality of an image so that important features are more visible. This is especially important when working with scanned documents or photos that may have different lighting conditions, colors, or other visual characteristics that could obscure the text. Here are some common techniques used for image enhancement:

1. **Grayscale Conversion**
2. **Histogram Equalization**
3. **Thresholding**
4. **Contrast Adjustment**

Image Restoration

The Image Restoration process aims to improve the appearance of an image by fixing problems like noise and blurriness. It uses mathematical models to understand how the image got damaged and applies techniques to clean it up. Common techniques involved in image restoration include Noise Reduction and Blur removal.

2.2.6.3 Object Recognition

Object recognition is the process of identifying and classifying different regions in an image, referred to as patterns or objects. There are two main approaches to object recognition

1. **Decision-Theoretic Approach:** This method uses numerical descriptions to analyze patterns. It looks at measurable features.
2. **Structural Approach:** This approach focuses on qualitative descriptions, using relational descriptors that describe the relationships and arrangements between different parts of the patterns.

2.2.7 Optical Character Recognition (OCR)

OCR or Optical Character Recognition is an input device used to read a printed text. OCR scans text in image, analyzes it character by character and converts it into machine-readable code and stores text on the system memory and processes it as a document. OCR is widely used in applications where transforming printed material into editable text.

2.2.8 OCR Architecture

After image processing, OCR comes into play. It extracts the text from the processed images, analyzing the document character by character to convert it into machine-readable code. It converts printed or handwritten text from forms into digital text, which can then be manipulated and stored. Once OCR extracts the raw text, text processing helps clean, format, and analyze this text.

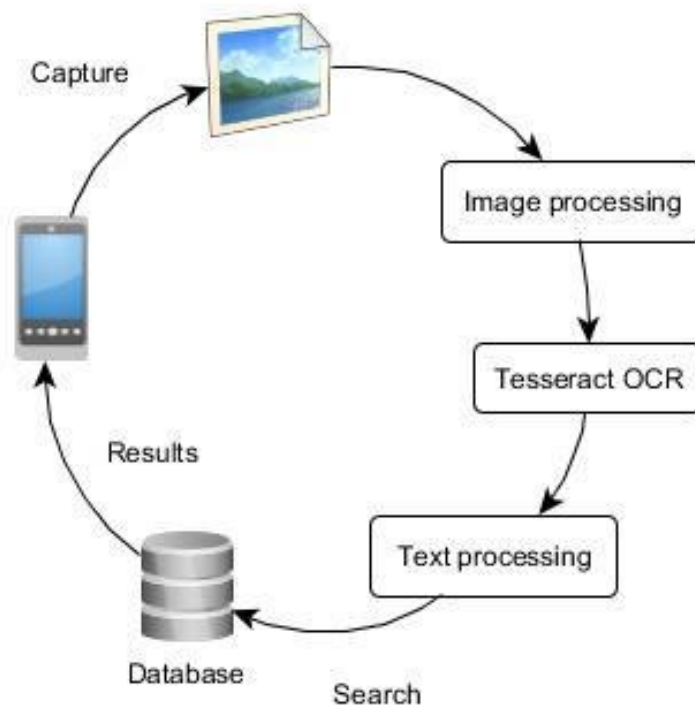


Figure 2.2 OCR Process from [researchgate.net](https://www.researchgate.net).

2.2.9 Portable Document Format

The Portable Document Format (PDF) is developed by Adobe in early 1993, it is a file format designed to display documents consistently across device and platform. The strength is preserve the formatting, fonts, images, and layout of the original document, ensuring a consistent and reliable presentation. A PDF file can include a varies type of content, such as text, image, vector graphics, hyperlinks, forms, annotations and even multimedia elements like audio and video.

2.2.9.1 PDF Categories

1. **PDF**: A standard PDF format, It is often used for sharing and viewing files online.
2. **PDF/A** : For long-term archiving; restricts elements like JavaScript and multimedia.
3. **PDF/E** : Tailored for engineering, construction, and manufacturing documents.
4. **PDF/X** : Preferred by print and graphic professionals for consistent visual output.
5. **PDF/VT** : Like PDF/X but allows variable and transactional printing (customization).
6. **PDF/UA** : Accessible format supporting assistive technologies for users with disabilities.
7. **PAdEs** : Standard for legally binding electronic signatures.
8. **PDF Healthcare** : Ensures best practices for healthcare-related data handling.
9. **Searchable PDF** : Combines image-based content with searchable text via OCR.

2.2.9.2 PDF File Structure

The PDF file consists of four part component that describe every pdf file, And all information in pdf is represent in ASCII

1. **Header:** The first line of the PDF file that specific the version of the PDF.
2. **Body :** Contains the document's main content, including pages, text, and images.
3. **Cross-reference table (XRef) :** Indexes the locations of all objects in the file.
4. **Trailer :** Provides structural information and indicates the starting point for reading the document.

2.2.10 Text Parser

Text parsing is the process of analyzing a string of text to extract meaningful information or convert the text into a structured format that a computer program can process. The primary goal of text parsing is to decompose and interpret text data, often transforming unstructured or semi-structured text into a structured form for further analysis or manipulation.

2.2.11 No Language Left Behind (NLLB)

No Language Left Behind (NLLB) is a groundbreaking AI project developed to address the challenge of multilingual translation, particularly for low-resource languages. The project aims to ensure that no language is left behind in digital communication by providing high-quality translation models for over 200 languages, including low-resource languages such as Asturian, Luganda, and Urdu. This initiative promotes linguistic diversity, enabling users around the world to access, share and communicate in their native languages. At its core, NLLB is a Multilingual Machine Translation (MMT) system that leverages deep learning techniques to train models capable of handling 218 languages across over 40,000 translation directions. The project introduces NLLB-200, a family of models with varying sizes and performance levels, allowing users to select models based on the computational resources available and the accuracy required for their applications.

2.2.11.1 Model Architecture and Training

NLLB-200 is built on the Transformer architecture, a deep learning model that has become the standard in natural language processing tasks. NLLB-200 employs several advanced techniques:

1. NLLB-200 uses a special design called a Mixture of Experts (MoE). Instead of using the full model for every input, it turns on just a few parts (called "experts") depending on the input sentence. This makes it possible to build a very large model up to 54 billion parameters without needing huge computing power all the time.
2. When translating, the model uses special tags to know which language it's translating from and to. It also uses one big shared list of words (called a vocabulary) across all languages. This helps the model understand patterns between different languages, even ones that are far apart.
3. A good translation model needs high-quality examples. Meta created tools to search and clean up sentence pairs in hundreds of languages. They used techniques like LASER3 (which helps find similar sentences in different languages) and other mining tools to gather the best data possible.
4. Before training the model to translate, Meta first let it read a huge amount of text in different languages. This helped the model learn how language works in general, even without translations. After that, they fine-tuned it using proper translation pairs..

2.2.12 Security of Web Form

Web form security is essential as these forms are highly vulnerable to cyberattacks, making them prime targets for malicious parties aiming to access sensitive user data. Common threats include:

1. **Cross-Site Scripting (XSS):** Attackers inject malicious scripts into web forms or application code, which can steal user information like keystrokes or cookies, or redirect users to harmful websites.
2. **SQL Injection:** Attackers manipulate SQL code within web forms to access or alter databases. This technique can expose or modify sensitive data, and it's one of the oldest and most dangerous web vulnerabilities.
3. **Cross-Site Request Forgery (CSRF):** Attackers trick users into performing unintended actions, such as submitting forms or transferring data, without their knowledge. These attacks exploit the user's browser and session data.
4. **Data Breaches:** A variety of attacks can lead to breaches, exposing sensitive customer information and damaging a business's reputation, as users lose trust in the website's security.

Many data privacy laws now regulate how web applications must ensure security, especially when handling personal information. The rules we must follow may depend on where our business is located and how it operates. To stay compliant, website owners need to adopt secure practices for web forms.

Key laws to be aware of

1. **GDPR (General Data Protection Regulation)**
2. **CCPA, CPRA (California Consumer Privacy Act and California Privacy Rights Act)**
3. **HIPAA (Health Insurance Portability and Accountability Act)**
4. **PCI DSS (Payment Card Industry Data Security Standard)**

Key measures for securing web forms

1. **Use TLS/SSL Certificates:** TLS/SSL certificates create encrypted links between browsers and servers, ensuring that data transferred is protected. Extended Validation (EV) certificates provide the highest level of security.
2. **Encrypt Data:** End-to-end encryption (E2EE) ensures that data is protected throughout its journey from sender to recipient, preventing unauthorized access. Use SSL certificates and CDNs to enhance security.
3. **Validate and Sanitize Input:** Validate and sanitize user inputs to prevent attacks like SQL injections or cross-site scripting (XSS). Validation checks if inputs are correct, while sanitization removes harmful data.
4. **Collect Minimal Data:** Only collect necessary information to reduce the impact of potential data breaches. This minimizes risk and aligns with privacy regulations like GDPR and CCPA.
5. **Anonymize Data:** Mask sensitive data (e.g., credit card numbers) using techniques like asterisks or data shuffling, making it harder for attackers to misuse it.
6. **Ask for Consent:** Obtain clear user consent before collecting personal data, and provide easy access to your privacy policy outlining how data is used and protected.

7. **Restrict File Uploads:** Limit file uploads by allowing only authorized users, setting file size limits, and using whitelist file types to prevent malware.
8. **Use reCAPTCHA:** Implement reCAPTCHA to differentiate between humans and bots, helping reduce spam and prevent automated attacks.
9. **Require Authentication:** For sensitive forms, require users to authenticate (e.g., through login or multi-factor authentication) to restrict access to authorized individuals only.
10. **Use Virus and Malware Protection:** Employ virus and malware protection through layered security, such as firewalls, malware scans, and intrusion detection, to safeguard your web forms from threats.

2.3 Languages and technologies

2.3.1 Web Development Language

2.3.1.1 HTML

HTML (HyperText Markup Language) is the standard language for creating and designing web pages. It is a markup language, which means that specific tags are used to structure web content. Each tag specifies a particular aspect of a website, such as text, photos, links, and multimedia.

2.3.1.2 TypeScript

TypeScript is a superset of JavaScript that introduces static typing. It was created by Microsoft to solve some of the limitations of JavaScript, particularly in large-scale applications. Because TypeScript is a superset, it supports all JavaScript features while also adding new functionalities, notably those aimed at increasing code dependability and maintenance.

2.3.2 Front-end Technology

2.3.2.1 Vue.js

Vue.js is a JavaScript framework for building user interfaces. It builds on top of standard HTML, CSS, and JavaScript and provides a declarative, component-based programming model that helps you efficiently develop user interfaces of any complexity.

2.3.2.2 TailwindCSS

Tailwind CSS is a utility-focused CSS framework that simplifies web development by providing pre-designed utility classes. These utility classes allow you to create custom designs.

2.3.2.3 Formkit

Formkit is modern form-building library for Vue.js, designed to help developers create and manage forms with minimal effort and maximum flexibility.

2.3.3 Backend Technology

2.3.3.1 FastAPI

FastAPI is a modern web framework for creating Python APIs that prioritizes speed, ease of development, and automated validation. It has gained popularity for developing quick, high-performance APIs because it uses Python type hints to provide explicit, automatic data validation and interactive API documentation.

2.3.3.2 Pydantic

Pydantic is a Python data validation and parsing library that enforces data types and restrictions using Python type annotations. Although it can be used separately, it is an essential part of FastAPI. With Pydantic models, you can easily serialize data, validate incoming data, and specify the structure of your data.

2.3.3.3 SQLAlchemy

SQLModel is the Python SQL toolkit and Object Relational Mapper that gives application developers the full power and flexibility of SQL without creating a SQL statement.

2.3.3.4 PostgreSQL

The PostgreSQL popular open-source relational database management system (RDBMS) for managing and storing data. It is renowned for being reliable, effective, and compliant with SQL standards. And It support storing a json format data type which it can be act as NoSQL in some situation.

2.3.4 Document Processing

2.3.4.1 Python

Python Programming language is a popular high-level computer programming language for machine learning because it is simple to use and easy to read and write. Additionally, it has a fast processing speed. Web development, data analysis, automation, scientific computing, and many more topics are among the many libraries available.

2.3.4.2 PyMuPDF

PyMuPDF is a A powerful Python package for extracting, analyzing, converting, and manipulating data from PDF documents.

2.3.4.3 NLLB-200

NLLB-200 (No Language Left Behind) is a project by Meta (Facebook) that focuses on creating AI models capable of translating between 200 languages.

2.3.4.4 PyTorch

PyTorch is an open-source deep learning framework developed by Facebook's AI Research lab (FAIR). It provides tools for building and training neural networks using dynamic computational graphs, which allow for flexibility and ease in model building.

2.3.4.5 LLAMA

LLama (short for Large Language Model Architecture) is a type of AI model designed to understand and generate human-like text. It is built using a transformer architecture, which helps it learn language patterns and produce meaningful responses based on its training data.

2.4 Competing solutions

2.4.1 Microsoft Azure Document Intelligence



Figure 2.3 Microsoft Azure Document Intelligence from [Learn.microsoft.com](https://learn.microsoft.com).

Microsoft Azure Document Intelligence is a solution that provides a feature of capturing data from printed or handwritten forms. And create a flow pipeline with Azure Cognitive Search pipeline to complete workflow as the user needs.

CHAPTER 3 DESIGN AND METHODOLOGY

This chapter will cover the features, architecture, functionalities, design methods, and diagrams of our web application. We will delve into the details of the application's functionality and architecture.

3.1 Project Functionality

3.1.1 System Requirements

- The web application must allow users to register, log in, and log out.
- The web application must allow the users to upload documents in formats of PDF and Image.
- The web application must allow users to fill the form without login.
- The web application must be able to identify a input field of the uploaded form.
- The web application must allow users to customize the form before publishing a access link.
- The web application must allow users to access the form via a access link to fill a information.
- The web application must allow users to view a result of the form.
- The web application must allow user to export a result as a CSV file.
- The web application must allow users to delete the form.

3.1.2 Feature List

3.1.2.1 Registration and Authentication System

The Registration and Authentication System must enable a user to register and login to the system by using only email and password. This including a forgot password and the OTP for reset password.

3.1.2.2 Form Layout Analysis Feature

The Form Layout Analysis System will extract all the text from the document that is uploaded by the user via a web application. The system will then analyze the form's pattern, extract the input labels, translate the text into English, and send the translated information to a generative AI for data type generation.

3.1.2.3 Form Schema generator Feature

The Form Schema generator System will receive information from the Paper-Based Form Analysis System and Form Schema generator System must be able to generate a form schema that compatible with a form library.

3.1.2.4 Web Application Form

The Web Form System must enable users to complete the form, store the data in the database, and see the data created by the form owner.

3.2 Use Case Diagram

From the figure 3.1 The diagram shows a relationship between the user and the system by using a use case diagram. The user of the system is a developer that needs to transform a paper-base form into web application form and the user who going to fill the form. The system consists of 3 different systems, which are form layout analysis system, form schema generator systems, web form systems and registration and authentication systems. The user can upload a paper-based form to the system, and the system will transform the form into a web-based form.

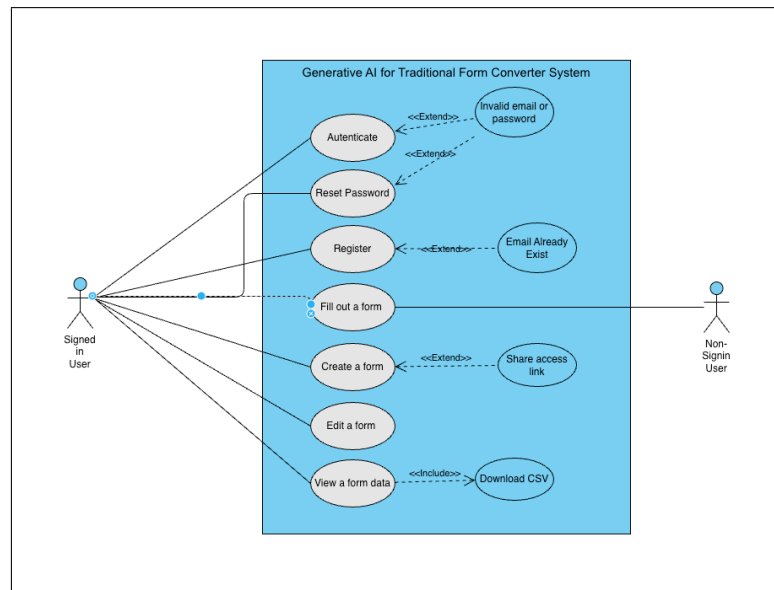


Figure 3.1 Use case diagram

3.3 Use Case Narrative

3.3.1 Authentication

Use Case Name: Authentication

Actors: Form Creator and Required Login User

Goal: Users log in to the system.

Preconditions: User is registered

Main Success Scenario:

1. User access the website.
2. User enter a email and password.
3. User submit a email and password.
4. System authenticate and navigate to the home page.

3.3.2 Register

Use Case Name: Register

Actors: Form Creator and Required Login User

Goal: Users register to a system.

Preconditions: User does not have an account

Main Success Scenario:

1. User access the website.
2. User click to create a new account.
3. User enter an email and password and personal information.
4. User submit information.
5. System saves user information and navigates users to the login page.

3.3.3 Create a form

Use Case Name: Create a form

Actors: Form Creator

Goal: Create a form by upload the form file.

Preconditions: User has logged in.

Main Success Scenario:

1. User go to home page.
2. User click at upload a form.
3. User select a file to upload.
4. System will process the file and navigate users to the edit form page to confirm a form before publishing.

3.3.4 Edit a form

Use Case Name: Edit a form

Actors: Form Creator

Goal: Edit a form to make a change.

Preconditions: User has logged in.

Main Success Scenario:

1. User go to home page.
2. User click at edit a form at the form user need to make change.
3. User make change a form.
4. User click back to the previous page.
5. System will process autosave and navigate users to the previous page.

3.3.5 View a form data

Use Case Name: View a form data

Actors: Form Creator

Goal: View the data that user have input

Preconditions: User has logged in.

Main Success Scenario:

1. User go to home page.
2. User click at view a data of the form.
3. User can view a form result.

3.3.6 Fill up the form

Use Case Name: Fill up the form

Actors: Required Login Users and Anonymous User

Goal: Add a new data to the form

Preconditions: User has logged in or non-login user.

Main Success Scenario:

1. User access a form via the public link
2. User fill up a form.
3. User submit a form data.
4. System will save the data and navigate to the form page again.

Alternate scenario (user access the form required a login without login):

1. User access a form via the public link
2. System will navigate to the login page After login completes the user will redirect back to the form page.

3.4 Activity Diagram

From Figure 3.2 The Activity diagram shows the sequence how Generative AI for Traditional Form Converter System is working.

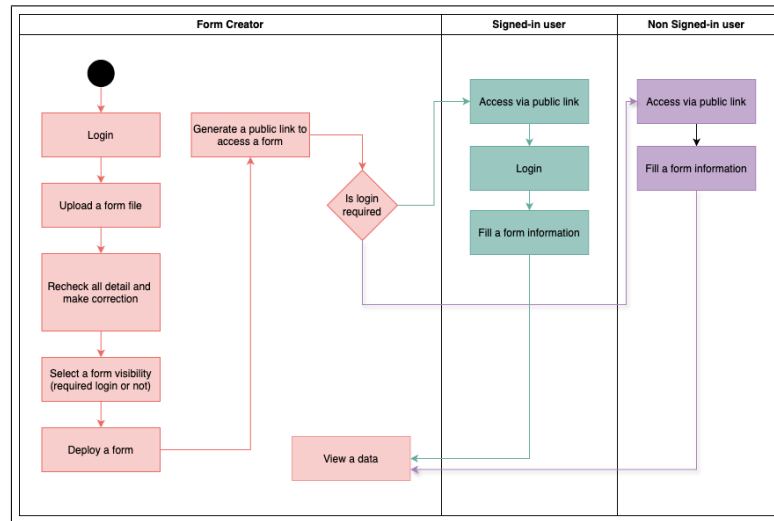


Figure 3.2 Activity Diagram

3.4.1 Form creator

- **Login:** Form Creator must login before use the system
- **Upload a form file:** After logging in, Form creator must upload a form file to add new form to the system.
- **Recheck all detail and make correction:** In this step, Form creator must check all the information that system has generated and make a correction if incase of error text found.
- **Select a from visibility:** Select a form visibility, whether the form creator need to form to be access by the user who signed-in or anyone can access.
- **Deploy a form and Generate a access link:** In this step, the form will be saved and generated a access link to allow user to access.

3.4.2 Signed-in user

- **Login:** If the form requires login, the user must log into the system.
- **Fill out information:** Once logged in, the user going to fills out the form and submit.

3.4.3 Non Signed-in user

- **Fill out information:** If login is not required, the user can directly fill out the form without logging in.

3.5 System Architecture

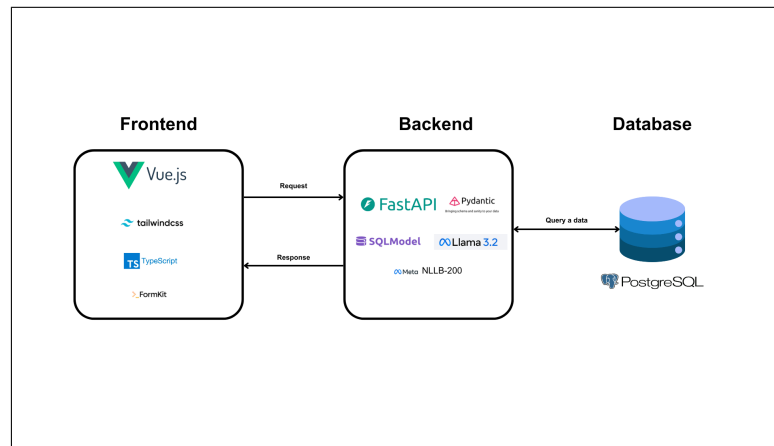


Figure 3.3 System Architecture

Figure 3.3, The diagram shown a system architecture in figure above, The System have divided into 3 part which front-end, back-end and database, each part have shown a technology stack, and here are the description of each component

- **VueJS** is a front-end Javascript library for building UI.
- **TailwindCSS** is a CSS framework for styling the UI and used with React.
- **FastAPI** is a Back-end framework for building REST API.
- **PostgreSQL** is a open-source relational database.
- **Pydantic** is a python library used for data validation.
- **SQLModel** is a Python base Object Relational Mapper (ORM) and SQL Tool kit.
- **Formkit** is a form engine library for Vue.js
- **NLLB-200** is a Machine learning Model for Text Translation from Meta.
- **PyMuPDF** is a high-performance Python library for data extraction, analysis, conversion and manipulation of PDF documents.
- **Llama 3:8b** is a text-based pre-trained and instruction-fine-tuned large language models.

3.6 System Workflows

3.6.1 Upload a file

In the upload a file step, the user will use a web application to upload a file into our system. The file types that the user can upload are PDF and image files. and we will forward the file to the following step once the user has uploaded it to the system.

3.6.2 Extracting text

Text extraction is the second step after a user uploads a file via the web application. In this step, a PDF library like PyMuPDF or pdfplumber is used to extract both the text and the layout from the PDF file. And for the image, we use Tesseract OCR to extract text and layout. After this step is completed, we will send a text to the text parser.

3.6.3 Text Parse

Text parsing is the third process after the extracted text has been completed. Text parse is the process of analyzing a text and breaking down the text to convert it into a structured format, In this process, the purpose of text parse is to analyze the layout of the form by using a text from a previous process to identify where which text is the label of the input field.

3.6.4 Translation

Translation is the fourth process and the first process that artificial intelligence have involve in the project. Translation is the process of translate a Thai input label to English Label. To make a Large Language identify a data type easily.

3.6.5 Data Type Generator

The Data Type Generator is the fifth process after preparing the process before the data type generator. The Data type generator is designed to generate a suitable datatype of each input field, for example, birthdate must be Date instead of string, passport number must be string, or text instead of number. All of these types will be applied with the form schema to make a form able to validate a correct data input.

3.6.6 Form Schema Generator

The Form Schema Generator is the last process of the workflow, the form schema generator will be used to generate a form schema by use the output received from the data type generator to show output on the webpage and interact with the user.

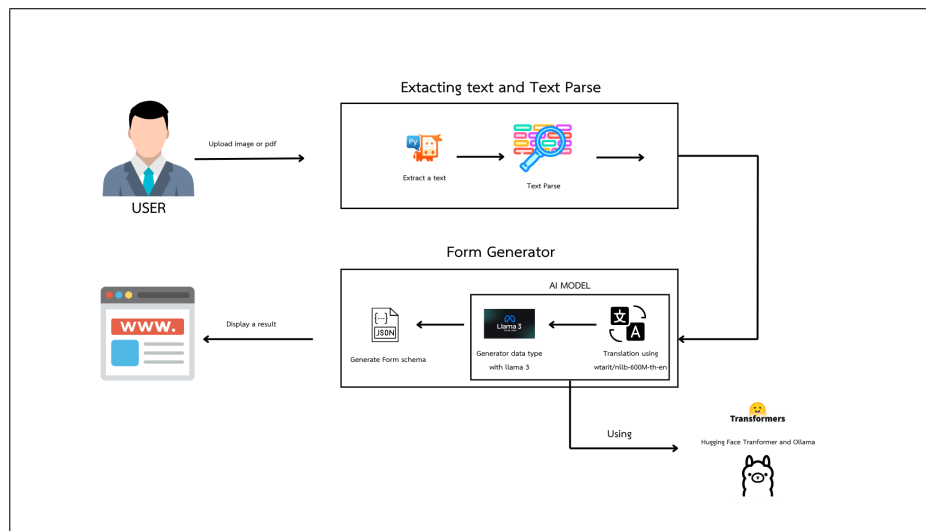


Figure 3.4 System Workflow Diagram

3.7 System Design

3.7.1 Text Extraction Design

The Text Extraction Design is responsible for transforming an image file or PDF file into machine-readable text for use in the text parsing process. The text extraction process will be divided into two primary components: text extraction from images and text extraction from PDFs. This step also involves text normalization,

which normalizes all dot styles to a standard dot. This is necessary because the text that has been detected will contain a variety of styles, including Unicode ellipsis, middle dot, and one dot leader.

3.7.1.1 PDF Text Extraction

PDF Text Extraction is using a python library called PyMuPDF to extract the text from and layout from PDF documents. This includes capturing elements such as paragraph positions, upper-case and lower-case text, which are essential for preserving the context and visual arrangement of the original document.

3.7.1.2 Optical Character Recognition Text Extraction

Our system also includes support for image-based form files using Optical Character Recognition, or OCR. This is especially useful when someone uploads a scanned document or a photo of a paper form instead of a digital PDF. Since text from images can't be read directly like from a PDF, OCR helps us turn the image content into readable and usable text. And The OCR process that we design is break down into three main steps.

Preprocessing

This is the first step before any actual text detection happens. We need to clean and prepare the image so the OCR tool can “see” the text more clearly. If an image is blurry, dark, or has weird formatting, it makes it harder for the tool to work correctly.

- **Grayscale Conversion:** We first turn the image into grayscale to make it simpler. Color doesn't really help with reading text, so removing it helps focus only on the shapes of letters.
- **Binarization (Thresholding):** Then we turn the grayscale image into just black and white. This makes the text stand out more by separating it from the background.
- **Noise Reduction:** Some scanned images have little dots or smudges that aren't part of the text. We try to clean those out so they don't confuse the OCR.
- **Deskewing** Sometimes an image isn't scanned straight, so the text is slanted. We try to fix that by straightening it, which helps keep the lines of text in order.
- **Rescaling and Contrast** For low-quality or blurry images, we may need to make the image bigger or increase the contrast so the letters are more visible.

Text Detection

Once the image is cleaned up, the next step is finding where the actual text is located in the image. We use an OCR engine called Tesseract, which looks for text blocks and figures out which parts of the image might contain useful text. It doesn't recognize the letters yet—it just locates the areas with text.

Character Recognition

After detecting where the text is, Tesseract tries to figure out each letter by comparing it to patterns it has already learned. It works for both English and Thai, which fits our project well. Once it's done, the output is plain text that we can use in the next steps of our form conversion process.

Limitations

Even though OCR is useful, it's not perfect. If the image quality is poor—like if it's too blurry, too dark, or has a weird font—the OCR might not read the text correctly. Also, handwritten text isn't supported. Because of these issues, we currently focus more on text-based PDFs for form extraction, but we still plan to improve the OCR feature in the future.

3.7.2 Text Parse Design

After extracting raw text and layout metadata from PDF documents and image, the text parsing is a processes that identify a form section and a form input label. This module is identify input style into 3 categories, which is a dot, underscore and dash. The Text Parse is designed to first identify the type of input label by counting the number of characters that match each input style. The input style with the highest number of characters will be used for the next process. After the previous process has identified the input style, the text parse will commence the process of identifying the input label based on the input style. This will be achieved by combining regular expressions and rule-based heuristics to create a common pattern that is compatible with the sample input that we have. The final step in this process is the output cleaning process. During this step, we will remove all non-alphabetic characters and trim any unnecessary space from the output. In Figure 3.5 will show a step of text parse.

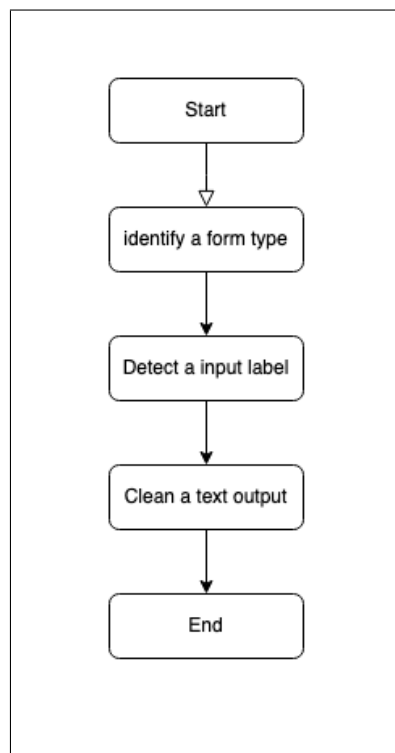


Figure 3.5 Text Parse Flowchart

3.7.3 Data Type Generator Design

In the Data Type Generator, we have designed to use a Llama as a large language Model (LLM) to determine the datatype of each input label. We use a structured prompting technique, which divides interactions with the model into two key components: a system prompt and a user prompt. The system prompt is used to set a behavior and role of the Llama, especially to instructing it to act as a data type classifier. After setting this context, the model receives a user prompt with the actual input label. It then analyzes the input and outputs the appropriate data type.

3.8 Database Design

Figure 3.6 shown a project database design, the system consist three tables in our project database design are user, form, and form result. table is used to store user data, form result is used to store a result that the user has filled out, and form table is used to store a form schema also a OTP and reset password session table.

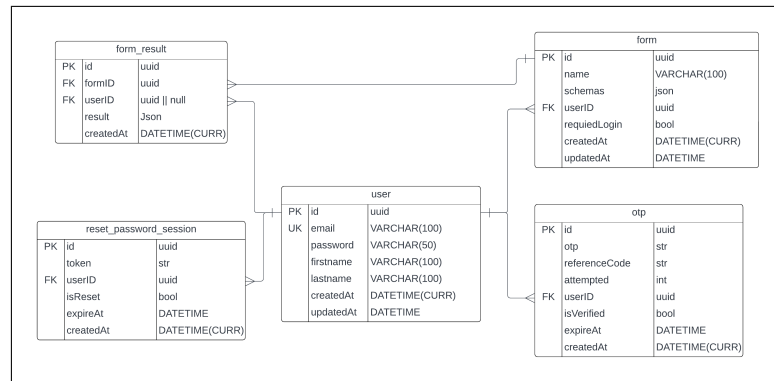
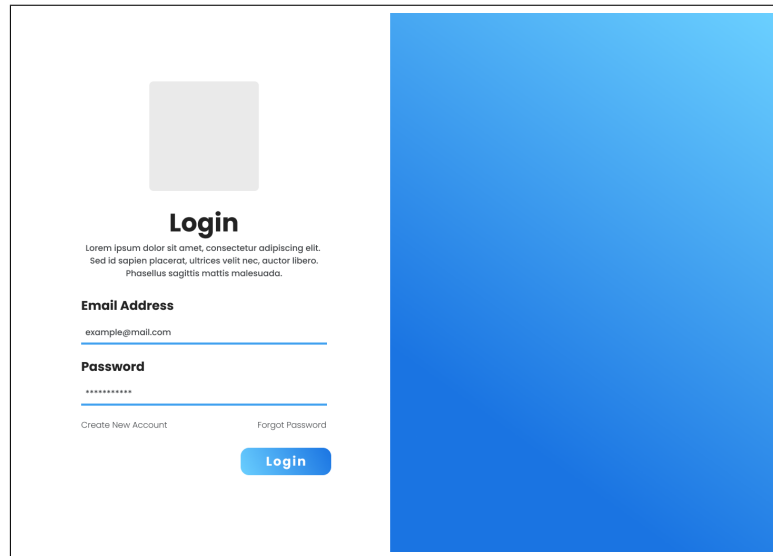


Figure 3.6 ER Diagram

3.9 User Interface Design

3.9.1 Login Page

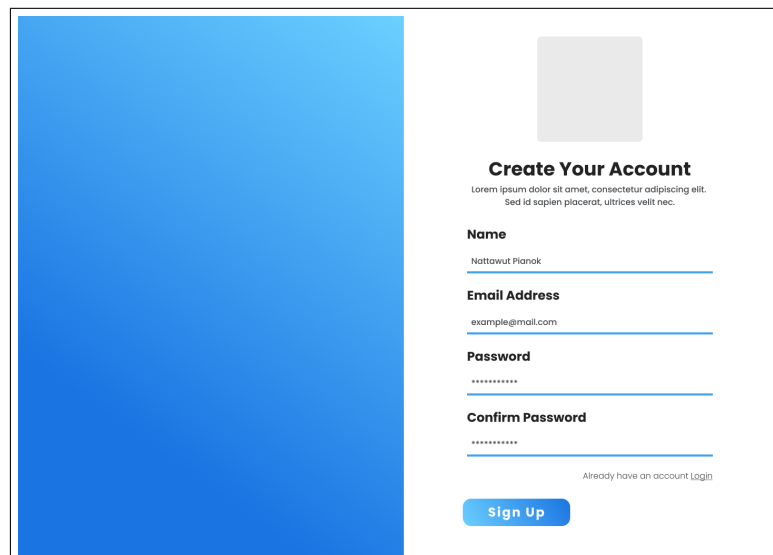


The login page features a white background on the left and a large blue gradient rectangle on the right. At the top left is a gray square placeholder for a profile picture. Below it, the word "Login" is centered in bold. Underneath is a paragraph of placeholder text: "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed id sapien placerat, ultrices velit nec, auctor libero. Phasellus sagittis mattis malesuada." Below this are two input fields: "Email Address" with the value "example@mail.com" and "Password" with masked characters "*****". At the bottom left are links for "Create New Account" and "Forgot Password". A blue "Login" button is centered at the bottom.

Figure 3.7 Login Page

Figure 3.7 represents the login screen of the web application. This page have a email field, password field and a login button to sent a credential to the back-end system.

3.9.2 Create Your Account



The create account page features a large blue gradient rectangle on the left and a white background on the right. At the top right is a gray square placeholder for a profile picture. Below it, the text "Create Your Account" is centered in bold. Underneath is a paragraph of placeholder text: "Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed id sapien placerat, ultrices velit nec." Below this are four input fields: "Name" with the value "Nattawut Pianok", "Email Address" with the value "example@mail.com", "Password" with masked characters "*****", and "Confirm Password" with masked characters "*****". At the bottom right is a link for "Already have an account Login" and a blue "Sign Up" button.

Figure 3.8 Create Account Page

Figure 3.8 represents the create account page of the web application. This page allow user to create their own account by the user must provide a following field which is name, email and password.

3.9.3 Edit Profile

Figure 3.9 represents the edit profile page of the web application. This page allow user to update their user profile by user can edit firstname, lastname and email.

Figure 3.9 Edit Profile Page

3.9.4 Forgot Password

At Figure 3.10 represents the create account page of the web application. When the user forgot their password, The user must navigate to this page by click at forget password from login page. And fill the email address to allow the system sent the One-time password (OTP) to email address.

Figure 3.10 Forgot Password Page

At the Figure 3.11, It represents a OTP confirmation page which required a OTP Code that the system have send to the email address. Figure 3.12, It represents a reset password pageIf it success the system will navigate to reset password page for enter a new password.

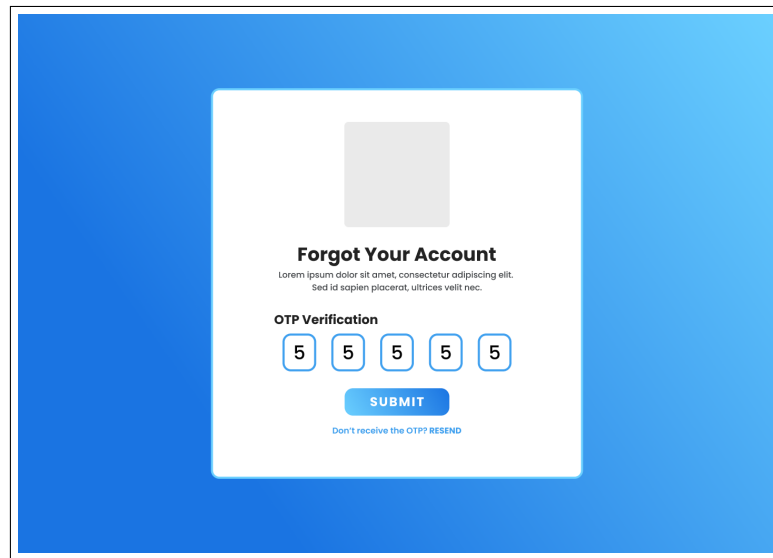
The image shows a 'Forgot Your Account' page. It features a light blue gradient background. In the center is a white card with a gray square placeholder for a profile picture. Below the placeholder is the title 'Forgot Your Account' in bold, followed by two lines of placeholder text. Underneath is the heading 'OTP Verification'. Below this heading are five blue square buttons, each containing the number '5'. Below the buttons is a blue 'SUBMIT' button. At the bottom of the card, there is a link that says 'Don't receive the OTP? RESEND'.

Figure 3.11 Forgot Password Page

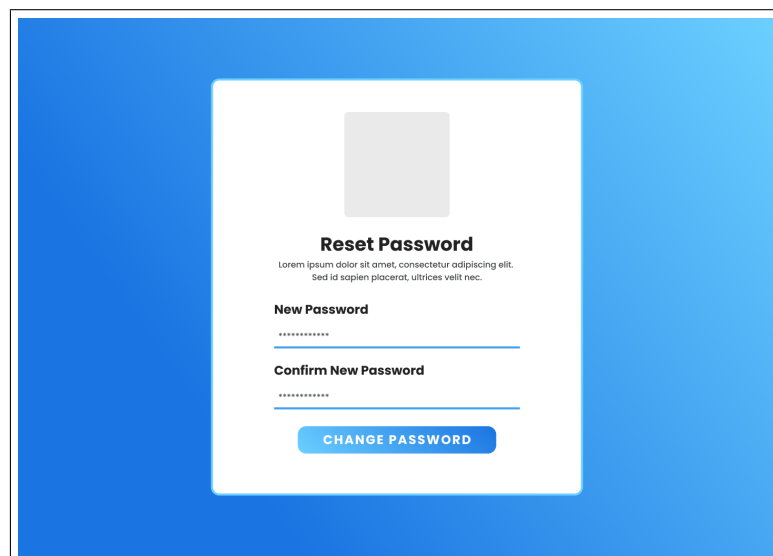
The image shows a 'Reset Password' page. It features a light blue gradient background. In the center is a white card with a gray square placeholder for a profile picture. Below the placeholder is the title 'Reset Password' in bold, followed by two lines of placeholder text. Underneath is the heading 'New Password'. Below this heading is a text input field with a blue underline. Below the input field is the heading 'Confirm New Password'. Below this heading is another text input field with a blue underline. At the bottom of the card is a blue 'CHANGE PASSWORD' button.

Figure 3.12 Reset Password Page

3.9.5 Dashboard

Figure 3.13 represents the dashboard page, it will show all the form that user have and the add form section at the left hand side, also have a upload status while file is uploading.

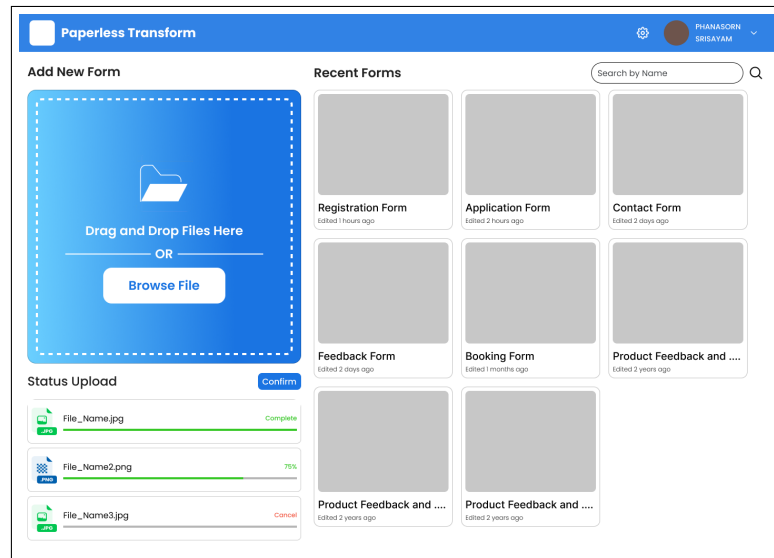


Figure 3.13 Dashboard Page

3.9.6 Edit Form

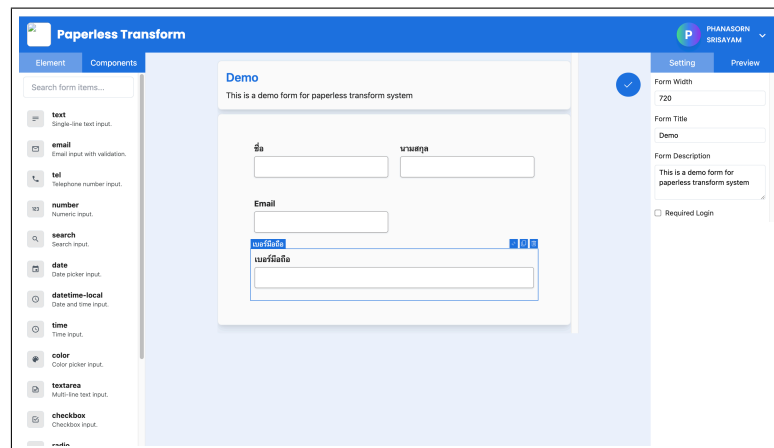


Figure 3.14 Edit Form Page

Figure 3.14 represents the Edit Form. The edit form page uses our own customize form editor based on Formkit, which allows users to fully customize a form, including changing input box length and add a input validation

3.9.7 Form Page

Figure 3.15 showcases the example of web form user interface, which it use a surveyJS to render the UI. This page will allow user to fill the information and submit a data to the system.

The image shows a web form titled "Deme Form" with a subtitle "This is a demo from for preview a paperless transform system". In the top right corner of the form area is a "Login" link. The form contains four input fields: "ชื่อจริง" (First Name), "นามสกุล" (Surname), "เบอร์โทรศัพท์" (Phone Number), and "Email". Below these fields is a blue "Submit" button. The entire form is centered on a blue gradient background.

Figure 3.15 Form Page

CHAPTER 4 IMPLEMENTATION RESULT

4.1 Introduction

This chapter will cover the entire working process and implementation progress of our project. Address complex issues as well. And we will divide each topic based on its features, which will include project data preparation.

4.2 Data Collection and Data Categorized

To study the structure of input fields and textual elements, approximately over 100 PDF forms were collected from various sources. These documents reflect a wide range of formatting styles and content arrangements, offering a diverse dataset for analysis. The findings from this analysis informed the development of methods for layout structure detection, text extraction, and input field identification. And we categorized the forms into three distinct styles based on input field presentation, which have dot-style forms, line-style forms, and tabular-style forms. And the following figure show the example of each form styles.

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แบบฟอร์มการแจ้งข้อร้องเรียนของลูกจ้างและการติดตามผล

1. วันที่ร้องเรียน...../...../.....
2. เรื่องที่ร้องเรียนเกี่ยวกับ ☐ สินค้า ☐ การบริการวิชาการ ☐ การบริการ ☐ บรรจุภัณฑ์
3. ข้อมูลผู้ร้องเรียน
ชื่อ – สกุล.....สังกัดหน่วยงาน.....
.....
โทรศัพท์.....
โทรสาร.....E-mail.....
4. รายละเอียดการร้องเรียน
 - 4.1 รายละเอียดของสินค้า หรือ บริการตามข้อตกลง / ใบยืนยัน
 - (1) เลขที่ใบส่งสินค้าและใบแจ้งหนี้.....
 - (2) วันที่รับสินค้า / รับบริการ.....
 - (3) รายการสินค้า / บริการ.....
.....
.....
.....
.....
 - 4.2 รายละเอียดของสินค้า หรือ บริการที่ร้องเรียน / ที่ไม่เป็นไปตามข้อตกลง
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ลงชื่อ.....ผู้ร้องเรียน / ผู้กรอกข้อร้องเรียนแทน
(.....)
วันที่.....

Figure 4.1 Example of dot-style forms

Medical Examination Form

(The Medical Examination will be conducted by any Govt. Gazatted Officer/Medical Officer at BGJ/H)

Items Nos. 1 to 8 below to be filled in by the candidate

1. Name of the candidate_____
2. Father's Name_____
3. Mother's Name_____
4. Date of Birth_____
5. Department (in which admission is being sought)_____
6. University Receipt for Medical Examination Fee
No._____ Date_____ Rs._____
7. Roll No. (allotted by the Department):_____
8. History of any previous or existing illness
 - I. History of illness like epilepsy, Hypertension, Asthma, Tuberculosis, Rheumatic, Arthritis, Diabetes, Heart Problem etc,
 - II. History of any Surgery / Accident_____
 - III. History of any medication_____

Photograph to be
attested by Physician

 (Signature of the candidate to be
Attested by the chairman)

 (Signature of the candidate in the
presence of the examining Doctor)

 (Signature of the chairman with seal
of the department)

Medical Examination

- A. General Physical Examination
 - a) Blood pressure
 - b) Pulse
 - c) Vision (without glasses) Right _____ left _____
 - d) Vision (with glasses) Right _____ left _____
- B. Laboratory Test
Urine : Alb _____
- C. Systemic Examination
- D. Any person specific recommendation requiring further tests / examination

It is certified that the above named candidate has been medically examined and found fit to pursue the course of studies to which he or she has already been admitted provisionally.

(Signature of the Medical Officer with seal and date)

Figure 4.2 Example of line-style forms

Form 1040 Department of the Treasury—Internal Revenue Service **2024** U.S. Individual Income Tax Return OMB No. 1545-0074 IRS Use Only—Do not write or staple in this space.

For the year Jan. 1-Dec. 31, 2024, or other tax year beginning . . . , 2024, ending . . . , 2024. See separate instructions.

Your first name and middle initial . . . Last name . . . Your social security number . . .

If joint return, spouse's first name and middle initial . . . Last name . . . Spouse's social security number . . .

Home address (number and street). If you have a P.O. box, see instructions. . . Apt. no. . . Presidential Election Campaign
City, town, or post office. If you have a foreign address, also complete spaces below. . . State . . . ZIP code . . . Check here if you, or your
Foreign country name . . . Foreign province/state/county . . . Foreign postal code . . . spouse if filing jointly, want \$3
to go to this fund. Checking a
box below will not change
your tax or refund. ☐ You ☐ Spouse

Filing Status ☐ Single ☐ Head of household (HOH) ☐ Married filing jointly (even if only one had income) ☐ Qualifying surviving spouse (QSS)
Check only one box. ☐ Married filing separately (MFS) If you checked the MFS box, enter the name of your spouse. If you checked the HOH or QSS box, enter the child's name if the
qualifying person is a child but not your dependent: . . .
☐ If treating a nonresident alien or dual-status alien spouse as a U.S. resident for the entire tax year, check the box and enter
their name (see instructions and attach statement if required): . . .

Digital Assets At any time during 2024, did you: (a) receive (as a reward, award, or payment for property or services); or (b) sell, exchange, or otherwise dispose of a digital asset (or a financial interest in a digital asset)? (See instructions.) ☐ Yes ☐ No

Standard Deduction Someone can claim: ☐ You as a dependent ☐ Your spouse as a dependent ☐ Spouse itemizes on a separate return or you were a dual-status alien

Age/Blindness You: ☐ Were born before January 2, 1960 ☐ Are blind **Spouse:** ☐ Was born before January 2, 1960 ☐ Is blind

Dependents (see instructions): (1) First name . . . Last name . . . (2) Social security number . . . (3) Relationship to you . . . (4) Check the box if qualifies for (see instructions):
Child tax credit ☐ Credit for other dependents ☐

If more than four dependents, see instructions and check here ☐

Income Attach Form(s) W-2 here. Also attach Forms W-2G and 1099-R if tax was withheld. If you did not get a Form W-2, see instructions.

1a Total amount from Form(s) W-2, box 1 (see instructions) . . . 1a
b Household employee wages not reported on Form(s) W-2 . . . 1b
c Tip income not reported on line 1a (see instructions) . . . 1c
d Medicaid waiver payments not reported on Form(s) W-2 (see instructions) . . . 1d
e Taxable dependent care benefits from Form 2441, line 26 . . . 1e
f Employer-provided adoption benefits from Form 8839, line 29 . . . 1f
g Wages from Form 8919, line 6 . . . 1g
h Other earned income (see instructions) . . . 1h
i Nontaxable combat pay election (see instructions) . . . 1i
z Add lines 1a through 1h . . . 1z

2a Tax-exempt interest . . . 2a b Taxable interest . . . 2b
3a Qualified dividends . . . 3a b Ordinary dividends . . . 3b
4a IRA distributions . . . 4a b Taxable amount . . . 4b
5a Pensions and annuities . . . 5a b Taxable amount . . . 5b
6a Social security benefits . . . 6a b Taxable amount . . . 6b

c If you elect to use the lump-sum election method, check here (see instructions) ☐

7 Capital gain or (loss). Attach Schedule D if required. If not required, check here ☐

8 Additional income from Schedule 1, line 10 . . . 8

9 Add lines 1z, 2b, 3b, 4b, 5b, 6b, 7, and 8. This is your **total income** . . . 9

10 Adjustments to income from Schedule 1, line 26 . . . 10

11 Subtract line 10 from line 9. This is your **adjusted gross income** . . . 11

12 **Standard deduction or itemized deductions** (from Schedule A) . . . 12

13 Qualified business income deduction from Form 8995 or Form 8995-A . . . 13

14 Add lines 12 and 13 . . . 14

15 Subtract line 14 from line 11. If zero or less, enter -0-. This is your **taxable income** . . . 15

For Disclosure, Privacy Act, and Paperwork Reduction Act Notice, see separate instructions. Cat. No. 11320B Form 1040 (2024)

Figure 4.3 Example of tabular-style forms

4.2.1 Data Processing

The Data Processing is a process of retrieving all the text from an uploaded file and convert to raw text content. For the image file has been implemented using Tesseract OCR, open-source library for character recognition from Google. it has been integrate with a image improvement and preparation system before process with a Tesseract, And in the Figure 4.4 show a result of the text after process with the OCR. And For the PDF file has been implemented using PyMuPDF, a high-performance library. It is a library that facilitates the extraction, analysis, conversion, and manipulation of PDF data. The text extraction from the PDF file is illustrated in Figure 4.5. However, there are still issues with the dot extraction process, as the dot character in some cases has multiple Unicode characters. To address this, we employ a text replacement algorithm to replace all potential dots with a standard dot or the Unicode character "U+002E" in order to normalize them. An additional issue is that a vowel in Thai text has been replaced with another character, such as "๑๐" when it should be "๑๐." Our research indicates that this issue is caused by the fact that the Unicode font is distinct from the standard font, and some may be the result of the compression of the PDF.

From Figure 4.4 the image shows a low accuracy of text extraction, with many Thai characters misrecognized, replaced by incorrect symbols, numbers, or English letters. Although some keywords and document structure are partially preserved, most of the content is unusable and severely garbled.

least 75 percentage of the corresponding word in the ground truth and is in the same position. This ensures that the evaluation not only checks for word similarity but also for correct placement in the sequence. The accuracy is then determined by the proportion of correctly matched words relative to the total number of words in the ground truth. Based on this method, the text parser achieved an accuracy of 82.9 percentage. Moreover, the text parser can't find a table layout, check box, or vector box because it only looks at text. which all of the following that have been mentioned can't be taken out as text.

```

Extracted Output:['เรียน', 'ข้าพเจ้า นาย/นาง/นางสาว', 'พนักงานตำแหน่ง', 'มีความประสงค์ขอลาออกจากการทำงาน', 'เนื่องด้วยเหตุผล', 'จึงขอสิ้นสุดการทำงาน ตั้งแต่วันที่', 'เดือน', 'พ.ศ']
Implemented Output:['เรียน', 'ข้าพเจ้า นาย/นาง/นางสาว', 'พนักงานตำแหน่ง', 'เนื่องด้วยเหตุผล', 'จึงขอสิ้นสุดการทำงาน ตั้งแต่วันที่', 'เดือน', 'พ.ศ', 'ขอแสดงความนับถือ']

```

Figure 4.6 Form Parse Result

4.2.3 Translation Thai to English

The Translation is use to translate a Thai language input label to english. we have use a NLLB-200 fined tuned model. Figure 4.7 below show that the translation is working correctly but may not it may get a text that we expect, such as วันเกิด expected value to be date of birth, but the predicted result show "Birthdate". which it have ability to translate into a correct meaning.

```

Excepted Output: Name, Firstname
Device set to use mps:0
Predicted Output: Name

Excepted Output: Lastname, Surname
Device set to use mps:0
Predicted Output: Last name

Excepted Output: Date
Device set to use mps:0
Predicted Output: Date

Excepted Output: Date of Birth
Device set to use mps:0
Predicted Output: Birthdate

```

Figure 4.7 Translation Result

4.2.4 Data Type Analyzer Result

As mentioned in Chapter 3, The data type analyzer is a process of generate a data type of each input label, by using a generative AI. And the result show that the data type analyzer can generate a correct and appropriate data type for each label. Importantly, the data type analyzer can output as a JSON array, as expected, enabling seamless integration into downstream processes.

Furthermore, in order to verify the accuracy of the analysis, we compared the generative AI type field with the proper types that humans (us) developed to serve as the ground truth, or the version that we anticipate the system to generate. The number of data types that matched the proper ones from the analyzer was then verified. After that, we converted the accuracy into a percentage by dividing the number of right answers by the total number of fields. The overall accuracy of the data type analyzer was 76.2 percentage.

Testcase Input	Expected Result	Result#1	Result#2	Summary
"name", "age", "district", "Birthdate", "postal_code"	{ "field": "name", "type": "text" }, { "field": "age", "type": "number" }, { "field": "district", "type": "text" }, { "field": "Birthdate", "type": "date" }, { "field": "postal_code", "type": "text" }	{ "field": "name", "type": "text" }, { "field": "age", "type": "number" }, { "field": "district", "type": "text" }, { "field": "Birthdate", "type": "date" }, { "field": "postal_code", "type": "text" }	{ "field": "name", "type": "text" }, { "field": "age", "type": "number" }, { "field": "district", "type": "text" }, { "field": "Birthdate", "type": "date" }, { "field": "postal_code", "type": "text" }	Passed✔
"firstname", "lastname", "age", "address", "phonenumber"	{ "field": "firstname", "type": "text" }, { "field": "lastname", "type": "text" }, { "field": "age", "type": "number" }, { "field": "address", "type": "text" }, { "field": "phonenumber", "type": "text" }	{ "field": "firstname", "type": "text" }, { "field": "lastname", "type": "text" }, { "field": "age", "type": "number" }, { "field": "address", "type": "text" }, { "field": "phonenumber", "type": "text" }	{ "field": "firstname", "type": "text" }, { "field": "lastname", "type": "text" }, { "field": "age", "type": "number" }, { "field": "address", "type": "text" }, { "field": "phonenumber", "type": "text" }	Passed✔
"ชื่อ", "นามสกุล", "อายุ", "ที่อยู่", "เบอร์โทรศัพท์"	{ "field": "ชื่อ", "type": "text" }, { "field": "นามสกุล", "type": "text" }, { "field": "อายุ", "type": "number" }, { "field": "ที่อยู่", "type": "text" }, { "field": "เบอร์โทรศัพท์", "type": "text" }	{ "field": "ชื่อ", "type": "text" }, { "field": "นามสกุล", "type": "text" }, { "field": "อายุ", "type": "number" }, { "field": "ที่อยู่", "type": "text" }, { "field": "เบอร์โทรศัพท์", "type": "text" }	{ "field": "ชื่อ", "type": "text" }, { "field": "นามสกุล", "type": "text" }, { "field": "อายุ", "type": "number" }, { "field": "ที่อยู่", "type": "text" }, { "field": "เบอร์โทรศัพท์", "type": "text" }	Failed✖ (Remark Only a phonenumber)

Figure 4.8 Data Type Analyzer Result

4.3 Web Application Implementation

The web application development is progressing well, with significant milestones achieved in the front-end implementation and back-end implementation. The following features have been successfully developed and tested.

4.3.1 Front-end development

In this project, we developed a frontend application using Vue 3 with TypeScript, styled with Tailwind CSS for a modern design. We utilized Pinia for state management, ensuring efficient and scalable data handling across components. Axios was integrated for HTTP requests, enabling seamless communication with the backend API. This setup provides a modular, maintainable, and performant user interface, supporting features such as authentication, form management, and user profile handling.

Authentication System: Includes Login, Sign-Up, and Forgot Password pages.

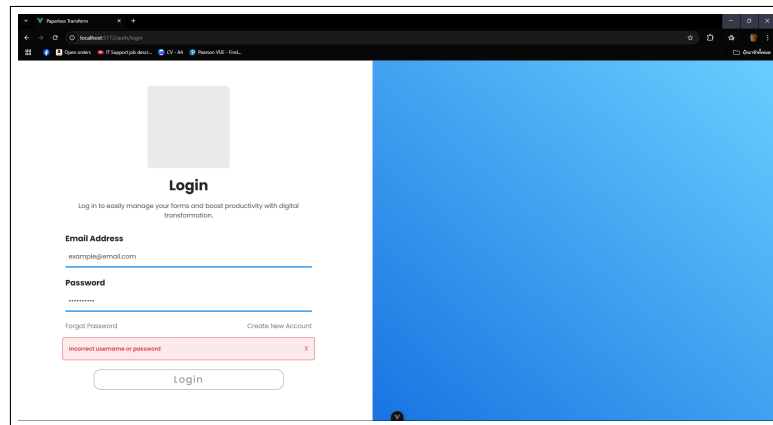


Figure 4.9 Implemented Login Page

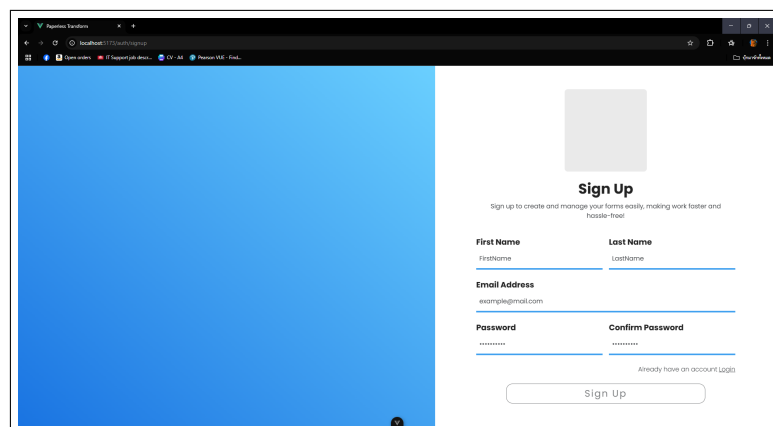


Figure 4.10 Implemented Signup Page

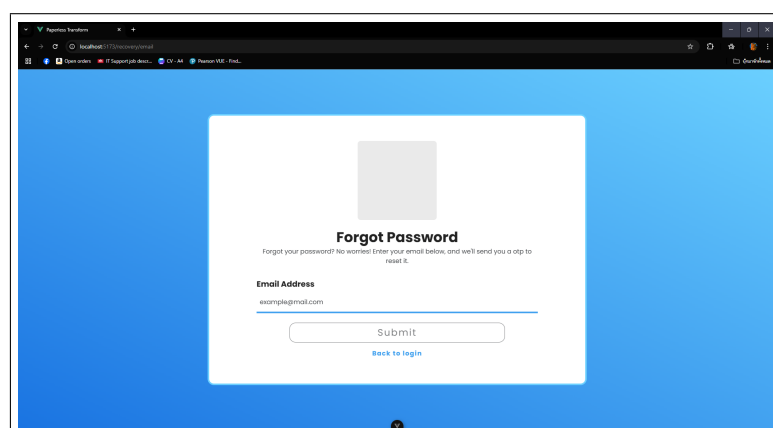


Figure 4.11 Implemented Forgot Passowrd

Dashboard Page: Dashboard Page: Fully implemented with necessary functionalities, such as upload files, status files upload, and recent form display with search by keyword.

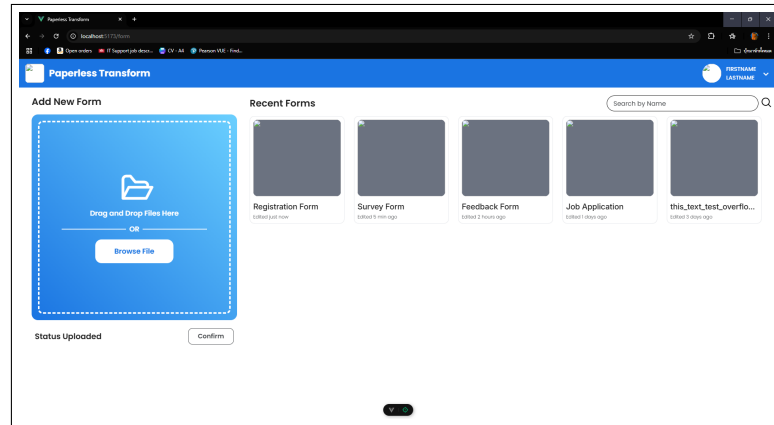


Figure 4.12 Implemented Dashboard Page

4.3.2 Back-end development

In this process, we construct a FastAPI application to manage a user and form within a PostgreSQL database. We have written an API by using class programming and utilized Pydantic to create a model to validate and support data from incoming requests. We also implemented a SQLAlchemy for object relation mapping to communicate with the PostgreSQL database. Additionally, we implemented a session management system with JWT.

The backend of the application is responsible for data manipulation, form management, and user authentication. It offers secure methods for user registration, login, password recovery, and profile management. Form creation, viewing, submission, and deletion are all supported by the API, which guarantees efficient data management. The routes are specifically designed to facilitate specific tasks, such as user authentication and form submission, while simultaneously guaranteeing data integrity and secure access. Efficient communication between the server and the frontend is facilitated by this infrastructure.

Table 4.1 Page Route API Specification

Route	Description	Method
/auth	The main authentication route that redirects users to the login page if they are not logged in. It is a guest-only route, preventing access for logged-in users.	GET
/auth/login	The login page where users can enter their credentials (email and password) to access their account.	GET
/auth/signup	The registration page where new users can create an account by providing necessary information (email, password, and other details).	GET
/recovery	Password recovery section, redirecting users to the email submission page. This section is designed for users who have forgotten their passwords.	GET
/recovery/email	The email submission form for password recovery. Users provide their registered email to initiate the password reset process.	GET
/recovery/otp	The OTP (One-Time Password) verification page where users enter the verification code sent to their email for password recovery.	GET
/recovery/reset	The password reset form where users can set a new password after successful OTP verification.	GET
/	The main dashboard route (for authenticated users only). It redirects to the form dashboard, serving as the home page for logged-in users.	GET
/form	The form dashboard page displays a list of all available forms that users can access, or manage.	GET
/form/result/:id	The form result page displays the submitted data for a specific form identified by the ID in the URL.	GET
/form/:id	The form display page renders a specific form identified by the ID in the URL, allowing users to fill and submit the form.	GET
/add	The form builder page allows users to create new forms by designing form fields and layout, supporting customization and editing.	GET
/profile	The profile settings page allows users to view and update their personal information, such as name, email, and password.	GET
/feedback	The feedback submission page enables users to provide feedback on the system, sharing their experience and suggestions for improvement.	GET
/aboutus	The About Us page provides information about the project, its purpose, and the development team behind it.	GET

Table 4.2 Authentication Routes API Specification

Route	Description	Method
/auth/signup	Registers a new user with the provided input for sign-up.	POST
/auth/signin	Authenticates a user with the provided input for sign-in.	POST
/auth/request-reset	Initiates the password reset process by requesting a reset.	POST
/auth/verify-otp	Verifies the OTP for password reset process.	POST
/auth/reset-password	Resets the user password with the provided new password input.	POST

Table 4.3 Form and User Routes

Route	Description	Method
/form	Fetches a list of all available forms, allowing the user to view multiple forms that are currently stored in the system.	GET
/form	Creates a new form using the input data provided by the user. This allows for the addition of new forms to the system.	POST
/form/:formId	Deletes a form identified by the provided formId from the system. This action removes the form and its associated data.	DELETE
/form/:formId/submit	Submits the form data for a specific form identified by the formId. This operation processes and stores the data submitted through the form.	POST
/form/:formId/result	Retrieves the result data for a specific form, identified by the formId. This allows users to view the results or output generated after the form was submitted.	GET
/user	Fetches the details of the currently authenticated user. This route returns information about the user's profile, preferences, or other relevant data.	GET
/user	Update user detail	PATCH

CHAPTER 5 CONCLUSIONS

5.1 Introduction

In this chapter, we will summarize the progress made on the project, detailing the work completed. We will discuss challenges encountered throughout the project, describing the issues faced and solutions adopted to address them. Finally, we will explore potential future work, identifying areas that could be improved to increase the overall effectiveness of the project.

5.2 Workloads

This project was carried out over the course of one academic year, divided into two semesters. In the first semester, after selecting the project topic, our team began by studying the fundamentals of document digitization, optical character recognition, form recognition, and text layout analysis. As none of the team members had prior experience in these areas, we focused on building a solid understanding of the core concepts and tools needed for converting PDF documents into structured web forms.

Following this, we researched existing frameworks and libraries that support PDF text extraction, form field detection, and table structure recognition. The team prioritized approaches that could handle diverse and complex form layouts to ensure flexibility in real-world applications.

During the second semester, we implemented the core system focusing on OCR and PDF to web form conversion. Key features included enhanced form layout detection and the upload file technique. The system was designed with a modular structure, allowing future extensions such as improved OCR integration for image-based documents and advanced table/form detection using machine learning. Additionally, we developed a web-based interface that enables users to upload documents, preview generated web forms, and manage conversion outputs efficiently.

Table 5.1 Workload and Progress Summary

Workloads	Progress
Research	
Learn about OCR	Completed
Learn about generative AI model	Completed
Learn about translation model	Completed
Learn about text parser	Completed
Learn about Python programming language	Completed
Learn about Node.js	Completed
Learn about Typescript	Completed
Design	
Design a use case diagram	Completed
Design a system workflow	Completed
Design a user interface	Completed
Implement	
Implement front-end	Completed
Implement back-end	Completed
Implement text extraction	Completed
Implement text parser	Completed
Implement translation	Completed
Implement Llama model	Completed
Test	
Manual testing	Completed
Deployment	
Deploy stability of server (front-end / back-end)	Completed

5.3 Problems and Solutions

5.3.1 Lack of Optical Character Recognition

From the outset, our objective was to create a system that could convert an image form into a web form. However, we encountered an issue with OCR accuracy, as the text that emerged from the OCR did not correspond to the text in the image. Nevertheless, the text parser is unable to extract a text label as a result of the OCR accuracy from the initial step. The text parser's main detection method is a dot, underscore, and dash, which are absent from the OCR-extracted text. To address this issue, we have decided to focus only a PDF file for transform into a web form and kept OCR for the future plan.

5.3.2 Lack of PDF Extraction

We have attempted to extract text from multiple forms and have observed that certain files contain vowels in the Thai language that have been replaced with numbers or special characters. The PDF itself displays the vowels correctly; however, when we extract the text using the library or manually copy the file, the result is an incorrect word containing a special character. In this instance, we conducted research and discovered that the vowel change was caused by a specific font, a special Unicode, or a compressed PDF file. To resolve this issue, we will verify the position of any unusual special text or number when the text is extracted. If it is identified, we will perform an error correction. However, this will be addressed in a future plan.

5.3.3 Lack of Tabular form support

One of the forms in the dataset we have gathered is tabular form style, which we are unable to support because the text parser we currently use only supports forms with lines and use a layout detection. We plan to support this in the future.

5.3.4 Language model for generating answers

Initially, we resolved to employ the open-source Llama 3.2:1B to perform the generating task in the design and methodology section. At first, we attempted to employ the open-source Llama 3.2; however, the output that was generated did not align with the instructions we provided. Resolution We have implemented a novel version of the generative AI, Llama 3:8B, which includes an eight billion parameter for training. The outcome has been satisfactory, as anticipated by the instructions we have provided.

5.4 Future Works

This project aims to develop a system for converting PDF documents into web forms to simplify data entry and form generation processes. The system focuses on transforming various PDF layouts—including those with complex tables—into structured web forms that can be easily filled and processed online. While the initial plan included optical character recognition (OCR) for extracting text from scanned documents, OCR integration has been postponed due to current accuracy limitations. Instead, the current phase prioritizes enhancing layout detection capabilities, supporting multi-file uploads, and preparing for future improvements in OCR and image-to-web form conversion. This approach lays the foundation for a more flexible and scalable document digitization workflow.

- Implement a form detection to detect more layout and support a tabular.
- Implement multi-file upload support, allowing users to upload several documents at once.

Although the OCR-based text extraction component, which was originally planned to support image-to-web form conversion, was not fully implemented due to current limitations in OCR accuracy, the system still delivers user-friendly functionality by focusing on reliable PDF-to-web form transformation. This missing feature has been noted for future development, once OCR performance improves.

Importantly, the system was designed with practical deployment in mind. It follows a modular architecture that allows integration into larger platforms, such as enterprise form management or document processing systems. The use of standard libraries, clear input/output interfaces, and support for multi-file uploads makes it well suited for real business scenarios, including bulk form digitization, administrative workflows, and data entry automation.

To further improve the performance, flexibility, and applicability of the system, the following areas are proposed for future development:

Form Detection and Tabular Support: Enhance the form detection component to recognize a wider variety of document layouts, including those containing complex structures such as tables and multi-column sections. This involves refining the layout analysis algorithms and integrating table structure recognition to ensure accurate mapping of PDF elements into corresponding web form fields. Techniques such as rule-based parsing combined with machine learning (e.g., layout classification or object detection models) can be applied to improve detection accuracy and robustness across diverse document types. This enhancement will enable the system to handle forms with more intricate designs, supporting a broader range of business and administrative use cases.

Checkbox Support: Implement functionality that allows users to upload multiple documents simultaneously, streamlining the workflow for processing large batches of files. This feature will support bulk conversion of PDF documents into web forms in a single operation, improving efficiency and user convenience. To ensure scalability and reliability, techniques such as asynchronous processing, file queue management, and progress tracking can be incorporated. This enhancement is particularly valuable in scenarios where users need to digitize large volumes of documents, such as in enterprise or government settings.

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